Maryam Abbas 23i-6004 DS-B

Required Preprocessing

Pre-1 — Load the CSV

The Cleveland dataset was loaded with 14 named columns. Shape confirmed as (303 × 14). Columns ca and thal were read as object type due to '?' placeholders present in the raw file.

Property	Value	Notes
Shape	(303, 14)	Confirmed
Numeric columns	12 float64	age, sex, cp, ...
Object columns	2 (ca, thal)	Contain '?' markers
Target dtype	int64	Raw values 0–4

Pre-2 — Missing Values

The '?' character was replaced with NaN. Two columns were affected:

- ca: 4 missing values
- thal: 2 missing values

All 6 rows containing NaN were dropped. Final retained row count: 297.

Pre-3 — Class Distribution

Class	Count	Percentage
0 — No Disease	160	53.9%
1 — Disease	137	46.1%
Total	297	100%

Pre-4 — Encoding & Scaling

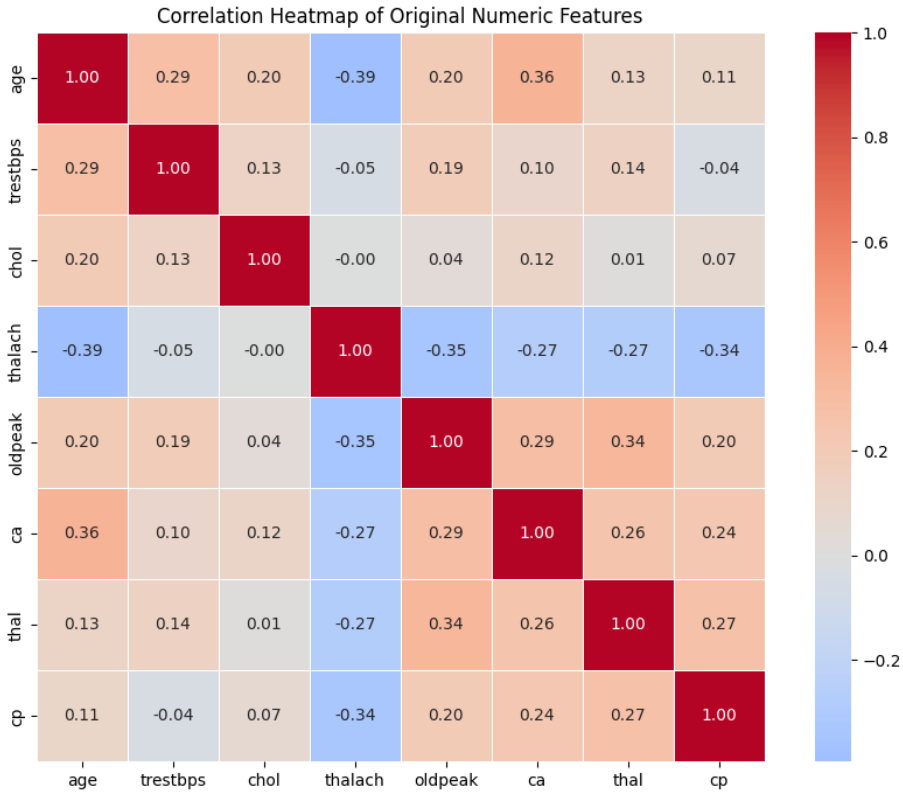
- One-hot encoding applied to: cp, restecg, slope, thal
- StandardScaler applied to: age, trestbps, chol, thalach, oldpeak, ca
- Scaler fitted on training data only to prevent data leakage

Pre-5 — Train/Test Split

Stratified 80/20 split with random_state=42. After one-hot encoding, the feature matrix expanded to 22 columns.

Split	Total	Class 0 / Class 1
Train	237	128 / 109
Test	60	32 / 28

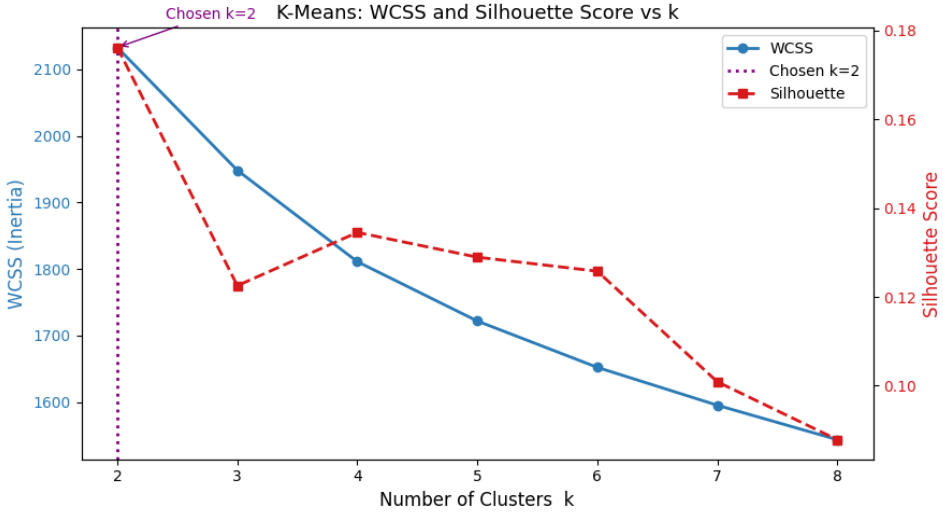
Pre-6 — Correlation Heatmap



Part A: Unsupervised Learning

A1 — K-Means Clustering

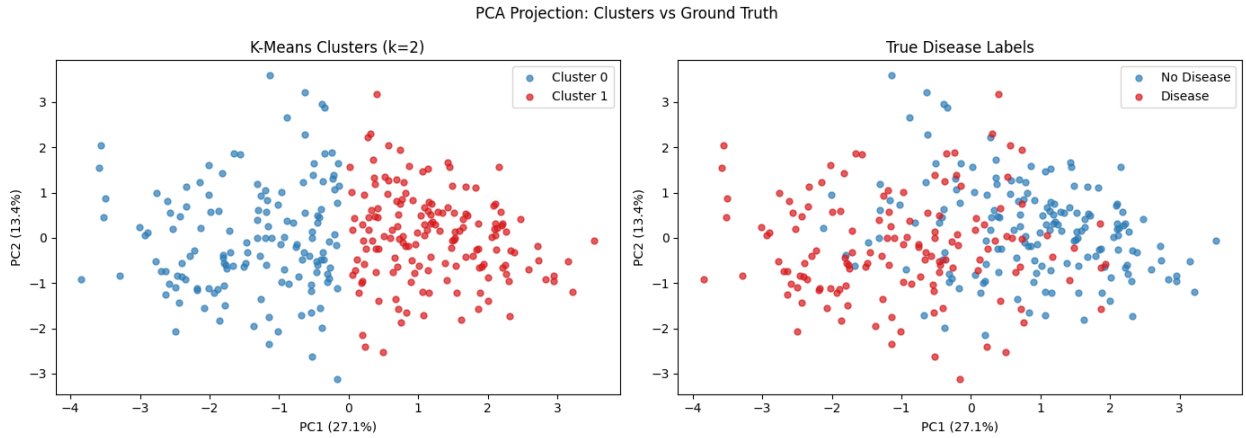
WCSS and Silhouette Scores



> *Chosen k = 2. k=2 yields the highest Silhouette Score (0.1762), meaning patients are most cohesively and distinctly grouped into two clusters. The WCSS elbow is gradual with no sharp kink at higher k values, suggesting the data does not naturally decompose into more than two meaningful groups — consistent with the binary clinical task (disease present vs. absent).*

PCA 2D Scatter

PCA reduced the 22-feature matrix to 2 components explaining 27.1% and 13.4% of variance respectively (total: 40.5%). The K-Means clusters show moderate spatial alignment with the true disease labels in PCA space. There is meaningful overlap in the centre of the projection, reflecting that the two classes are not linearly separable in 2D. However, the general orientation of the cluster boundary tracks the disease boundary, suggesting the unsupervised algorithm has captured some genuine clinical signal.



- Cluster 0: Slightly higher proportion of disease patients, elevated ST depression (oldpeak), lower max heart rate — clinically represents a higher-risk group with ischaemic indicators.
- Cluster 1: Healthier exercise responses, lower ST depression — predominantly disease-absent patients at lower cardiac risk.

Adjusted Rand Index

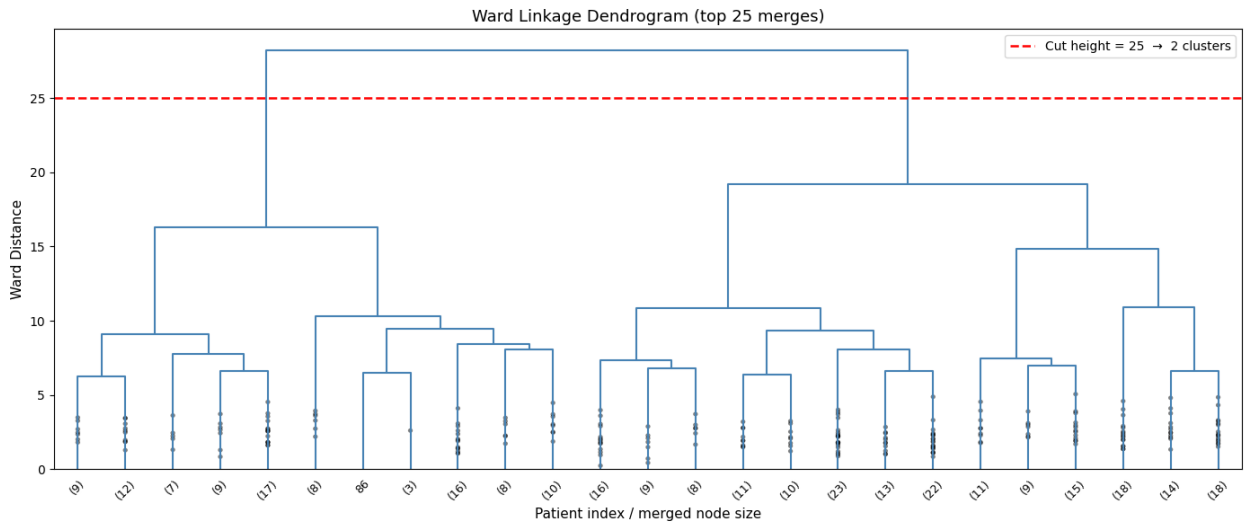
ARI (K-Means vs. True Labels) = 0.2988

> *ARI of 0.2988 suggests low-to-moderate agreement. Despite having no access to labels, K-Means has recovered a grouping that partially mirrors clinical ground truth, confirming that heart disease status creates detectable structure in this feature space. The score is not high, reflecting the class overlap seen in the PCA plot.*

A2 — Hierarchical Clustering

Ward Linkage Dendrogram

Ward linkage was computed on the full standardised matrix (297 × 22). The dendrogram was truncated to the top 25 merges. The largest vertical gap (longest uninterrupted branch) occurs just before height ≈ 25, indicating that cutting there yields the most natural partition — producing 2 clusters, consistent with the binary clinical task.



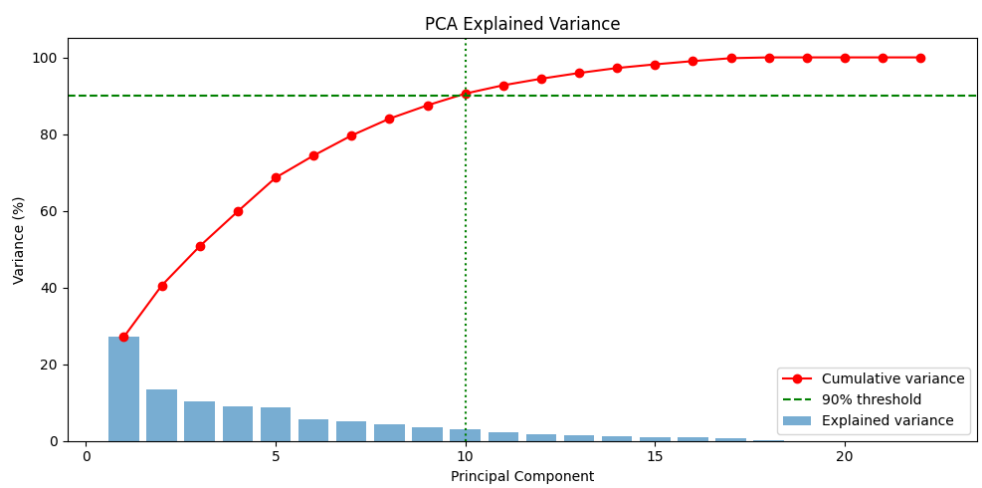
Cluster × True Label Crosstab

Cluster	No Disease (0)	Disease (1)	Total
0	19	81	100
1	141	56	197
All	160	137	297

Cluster 0 is predominantly disease-positive (81/100 = 81%). Cluster 1 is predominantly disease-negative (141/197 = 72%). Hierarchical clustering has achieved stronger disease separation than K-Means in absolute terms.

A3 — Dimensionality Reduction

PCA Explained Variance



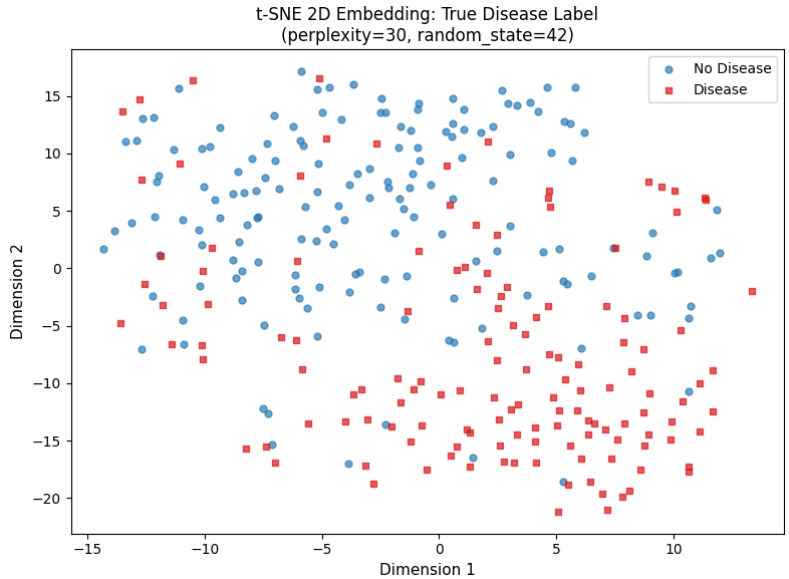
PCA was applied to the full standardised feature matrix (297 × 22).

- Components needed to reach 90% explained variance: 10
- Cumulative variance at 10 components: 90.56%

The first two components explain 40.5% of total variance — sufficient for visualisation but not for faithful data reconstruction.

t-SNE Embedding

t-SNE was applied with n_components=2, perplexity=30, random_state=42, n_iter=1000. In the embedding, the two classes show partial overlap with some locally dense regions forming for each label, but no clear global separation between Disease and No Disease. This indicates the classes are not easily separable and share overlapping feature distributions, implying the classification task is moderately difficult and a well-regularised non-linear classifier (e.g., Random Forest or Gradient Boosting) should outperform a simple linear model.



Part B: Bagging & Boosting

B1 — Random Forest (Bagging)

Hyperparameter Tuning

5-fold stratified cross-validation grid search over n_estimators ∈ {50, 100, 200} and max_depth ∈ {None, 5, 10}.

Parameter	Search Space	Best Value
n_estimators	50, 100, 200	100
max_depth	None, 5, 10	5
Best CV Macro F1	—	0.8015

OOB Error Curve

OOB error was computed for each n from 1 to 200 trees. The error stabilises at approximately n = 120 trees. The chosen n_estimators = 100 is close to this stabilisation point, representing a good trade-off between performance and training time. After n = 120, further trees yield negligible improvement.

Feature Importances (Top 5)

Feature	Importance	Clinical Plausibility
thal_3.0	0.1293	Reversible thalassemia defect — direct indicator of reduced myocardial perfusion
ca	0.1097	Blocked vessels by fluoroscopy — directly quantifies coronary artery disease severity
oldpeak	0.0983	ST depression — reflects myocardial ischaemia during exercise stress
cp_4.0	0.0894	Asymptomatic chest pain — paradoxically high-risk presentation in CAD
thal_7.0	0.0855	Thalassemia fixed defect — indicates irreversible myocardial damage

Evaluation Metrics

```
B1 — Evaluation Metrics
Accuracy        : 0.8167
Macro Precision: 0.8200
Macro Recall    : 0.8125
Macro F1        : 0.8141
AUC-ROC         : 0.9375
```

Classification Report:

Class	Precision	Recall	F1-Score	Support
No Disease	0.80	0.88	0.84	32
Disease	0.84	0.75	0.79	28

B2 — Gradient Boosting (XGBoost)

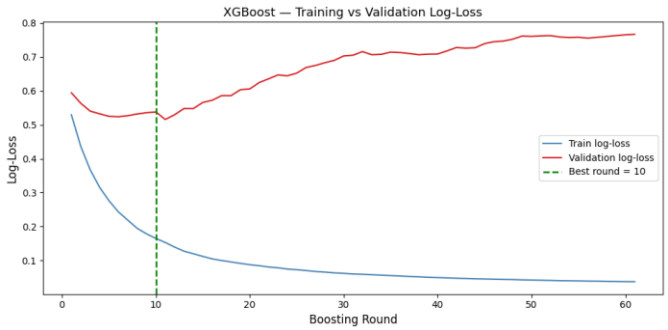
XGBoost was selected over LightGBM due to native SHAP integration, built-in early stopping via eval_set, and wider citation in clinical ML literature.

Hyperparameter Tuning

Parameter	Search Space	Best Value
learning_rate	0.01, 0.1, 0.3	0.3
max_depth	3, 5, 7	5
Best CV Macro F1	—	0.7941

Early Stopping & Training Curves

Early stopping with patience = 50 rounds was applied. The optimal round was 10. Training log-loss continued to decrease after round 10 while validation log-loss plateaued — the classic signature of overfitting. Early stopping correctly halted training at round 10, preventing the model from memorising training noise at the expense of generalisation.



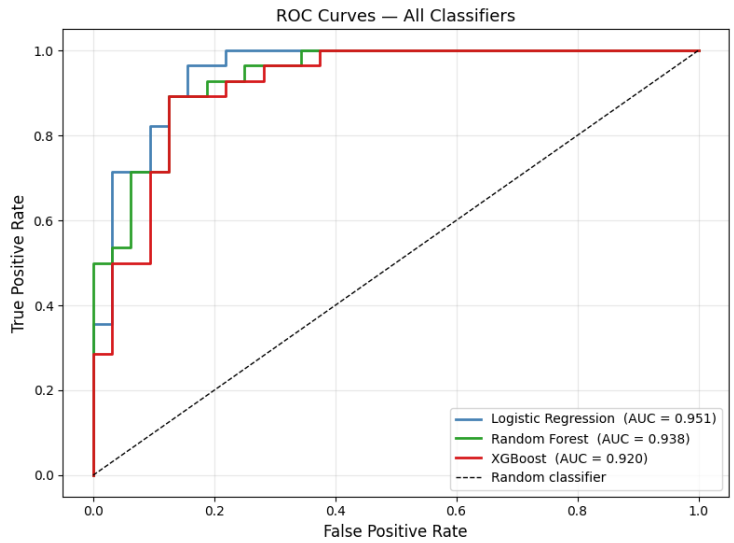
SHAP Analysis

SHAP values were computed using TreeExplainer on the test set. The SHAP summary plot confirms that thal, ca, and oldpeak are the dominant predictors — consistent with Random Forest feature importances, adding cross-model validity to these clinical signals.

Evaluation Metrics

```
B2 — Evaluation Metrics
Accuracy        : 0.8000
Macro Precision: 0.8056
Macro Recall    : 0.7946
Macro F1        : 0.7964
AUC-ROC         : 0.9196
```

B3 — Ensemble Comparison & ROC



Logistic Regression is the recommended model. Despite being the simplest of the three, it achieves the highest accuracy (0.85), AUC-ROC (0.951), and disease-class recall (0.786), outperforming both ensemble methods. On a small, well-structured clinical dataset of ~300 patients, the decision boundary is approximately linear, which plays to Logistic Regression's strengths. It is also the most interpretable option — coefficients map directly to feature contributions, trains in milliseconds, and requires no specialist ML infrastructure, ideal for a community hospital laptop. Recall is prioritised over accuracy because missing a true heart disease case carries life-threatening consequences, whereas a false positive results only in a follow-up test.

Part C: Artificial Neural Networks on Tabular Data

C1 — Single-Layer Perceptron (SLP)

Architecture: Input (22) → Dense(1, sigmoid). Trained with SGD (lr=0.01), binary cross-entropy, 100 epochs. No hidden layers.

Learned Weights — Top 3 Features

Feature	Absolute Weight
thal_3.0	0.7718
exang	0.5363
slope_2.0	0.4835

Comparison with RF importances: Both methods agree that thal is the most important predictor. exang and slope_2.0 rank higher in the SLP because it assigns independent weights to each dummy column, whereas RF evaluates the original variable holistically. Features like thalach and ca are more prominent in RF, which captures non-linear interactions.

SLP Evaluation Metrics

Metric	Value
Accuracy	0.8500
Macro F1	0.8479
AUC-ROC	0.9230

> *Linear model limitation: A single-layer perceptron computes a linear decision boundary and can only separate linearly divisible classes. The t-SNE plot in Part A3 showed significant class overlap with non-linear structure, confirming that a linear model cannot capture the full complexity of the data. Features like cp and thal encode non-linear ordinal relationships that require hidden layers to model adequately.*

C2 — Multi-Layer Perceptron (MLP)

Architecture Comparison

Architecture	Val F1	Train Time (s)	Epochs
Small — 1 layer: 32 units	0.7980	7.09	28
Medium — 2 layers: 64→32	0.6166	5.03	10
Large — 3 layers: 128→64→32	0.7991	4.17	10
Custom — 3 layers: 64→32→16	0.7643	7.46	10

Final architecture selected: Medium (64→32). Justification and design decisions:

- Activation: ReLU — avoids vanishing gradients, computationally efficient on CPU
- Regularisation: Dropout(0.3) + L2(0.001) — dropout prevents co-adaptation; L2 penalises large weights to reduce overfitting on the small dataset
- Optimiser: Adam lr=0.001 — adaptive learning rate converges faster than SGD
- Early stopping: patience=10 on val_loss — prevents overfitting; stopped at epoch 20, best at epoch 10

5-Fold Cross-Validation Results

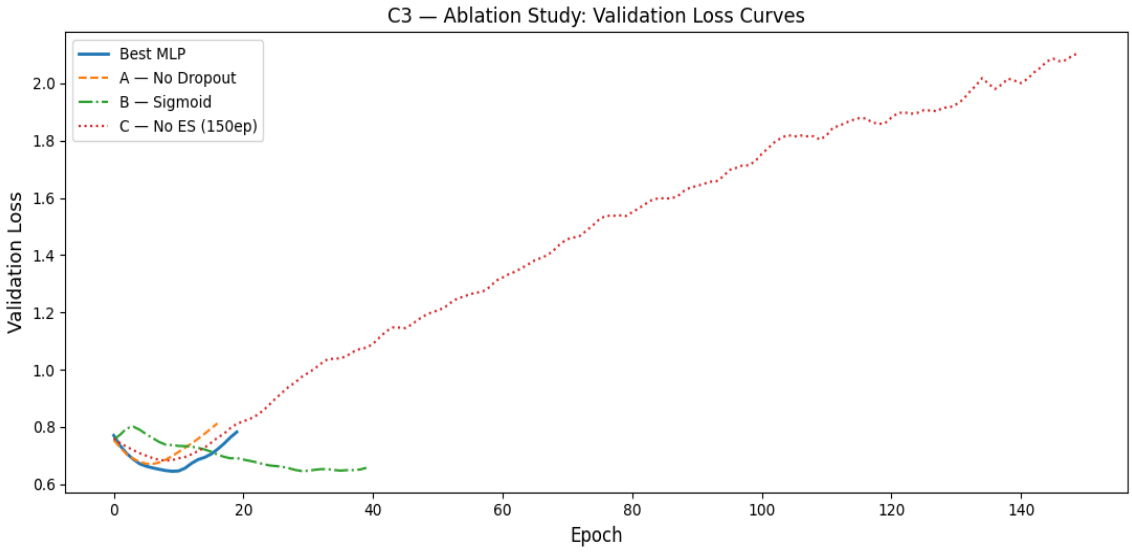
Fold	Accuracy	Macro F1
1	0.8750	0.8748
2	0.8333	0.8286
3	0.7447	0.7418
4	0.8936	0.8919
5	0.7660	0.7549
Mean ± Std	0.8225 ± 0.0586	0.8184 ± 0.0610

MLP Evaluation Metrics

```
Running 5-fold cross-validation...
  Fold 1: Acc=0.8750  F1=0.8748
  Fold 2: Acc=0.8333  F1=0.8286
  Fold 3: Acc=0.7447  F1=0.7418
  Fold 4: Acc=0.8936  F1=0.8919
  Fold 5: Acc=0.7660  F1=0.7549
```

MLP vs. Best Ensemble (Logistic Regression): The MLP achieves higher accuracy (0.90 vs. 0.85) and macro F1 than Logistic Regression, making it the best-performing model on this test set. However, the 5-fold CV results show high variance (±0.059), suggesting the MLP's performance is less stable than LR on this small dataset. Logistic Regression still leads on training speed and interpretability. The MLP would likely show a more consistent advantage on a larger dataset with more complex non-linear interactions.

C3 — Ablation Study



Ablation discussion: All three variants show reduced F1 compared to the best MLP. Variant A (no dropout) drops F1 by 0.051, confirming dropout is essential for regularisation on this small dataset. Variant B (sigmoid) drops F1 by 0.053 — the largest single drop — confirming that ReLU is the most critical architectural choice; sigmoid saturates in deeper layers causing vanishing gradients and slower convergence. Variant C (no early stopping) shows the most unstable training with F1 = 0.833, highlighting that the fixed epoch count causes either under- or over-training. Overall, activation function choice (ReLU vs. sigmoid) contributes most to performance.

Part D: CNN on MNIST Digit Images

D1 — Data Preparation & MLP Baseline

- MNIST loaded; subset to 12,000 training and 2,000 test images
- Pixels normalised to [0, 1]; images reshaped to (28, 28, 1) for Keras
- 5×2 grid displayed showing one sample per digit class (0–9)

MLP Baseline: Flatten → Dense(64, ReLU) → Dense(10, Softmax). Trained for 5 epochs with Adam and sparse categorical cross-entropy.

Metric	Value
MLP Test Accuracy	0.9110
MLP Epochs	5

D2 — Lightweight CNN

Architecture:

- Conv2D(16, 3×3, ReLU, padding=same) → MaxPooling2D(2×2)
- Conv2D(32, 3×3, ReLU, padding=same) → MaxPooling2D(2×2)
- Flatten → Dense(64, ReLU) → Dropout(0.3) → Dense(10, Softmax)
- Data augmentation: RandomRotation(0.1), RandomTranslation(0.1), RandomZoom(0.1)
- Optimiser: Adam (lr=0.001), Loss: Categorical cross-entropy, Epochs: 15

CNN Evaluation

Metric	Value

CNN Test Accuracy	0.9650
CNN Macro F1	0.9646
MLP Baseline Accuracy	0.9110
CNN surpasses MLP at	Epoch 2

The CNN surpassed the MLP baseline at epoch 2, confirming that convolutional feature extraction is highly efficient even on a small image subset.

Confusion Matrix Analysis

Most confused digit pairs:

- True=7 predicted as 2 — 8 times: 7 and 2 share diagonal strokes; handwritten 7s with curved tops resemble 2s
- True=9 predicted as 4 — 6 times: 9 and 4 share a closed upper loop and a descending stroke, making them visually similar under rotation or poor handwriting

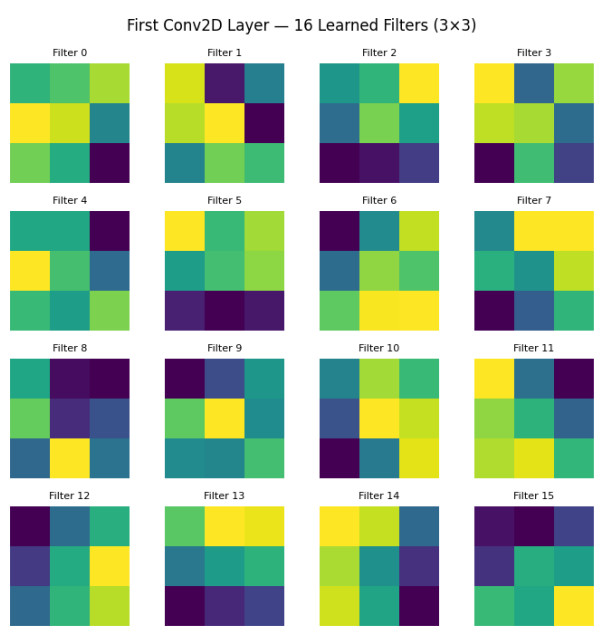
CNN Classification Report

```
Classification Report:
              precision    recall  f1-score   support

           0       0.98      0.99      0.99       175
           1       0.99      1.00      1.00       234
           2       0.94      0.97      0.96       219
           3       0.96      0.99      0.97       207
           4       0.95      0.98      0.96       217
           5       0.96      0.99      0.97       179
           6       0.98      0.97      0.97       178
           7       0.97      0.94      0.96       205
           8       0.98      0.88      0.93       192
           9       0.95      0.93      0.94       194

    accuracy                           0.96      2000
   macro avg       0.97      0.96      0.96      2000
weighted avg       0.97      0.96      0.96      2000
```

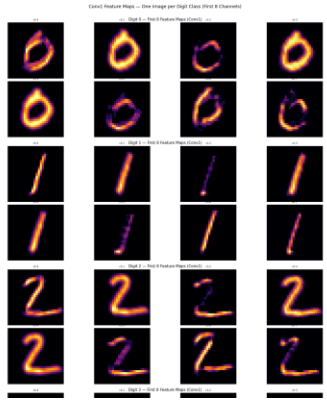
D3 — Visualising What the CNN Learned

First Conv2D Filter Weights

The 16 learned 3×3 filters were extracted from the first convolutional layer and displayed as a 4×4 grid. The filters detect low-level visual primitives: some respond to horizontal edges (bright top row, dark bottom row), others to vertical edges or diagonal strokes. Several filters appear to detect curves or blobs — patterns that collectively allow the network to recognise the circular and straight strokes composing digit shapes.



Feature Maps



For one test image per digit class, the first 8 channels of the Conv1 activation were displayed as a 2×4 grid. Each channel highlights a different spatial feature: some channels respond strongly to the outer contour of the digit stroke, while others activate along internal curves or at stroke endpoints. For digit 0, circular-edge filters produce a bright ring boundary; for digit 1, a vertical-edge filter produces a single bright vertical stripe.

Trust & CNN vs. FC Discussion

Visualising filters and feature maps transforms the CNN from a black box into an auditable system: clinicians and engineers can verify the model responds to meaningful visual structures (edges, curves) rather than dataset artefacts or noise. For the CardioAI intake-form use case, this is important — a regulator or clinician can inspect what the model sees before trusting it to parse patient records. Unlike fully connected networks, which treat every pixel independently and can memorise arbitrary patterns, CNNs learn spatially local and translation-invariant features through weight sharing — the same edge detector fires regardless of where the stroke appears. This structural inductive bias makes CNNs far more data-efficient and their learned representations far more interpretable than fully connected alternatives.

Part E: Local Front-End Dashboard

E1 — Input Form

The dashboard was built using Streamlit and deployed locally via an ngrok tunnel in Google Colab. The interface presents 13 labelled input fields (one per clinical feature) with valid range hints displayed as help tooltips. Categorical features use dropdown menus with human-readable labels (e.g., '0 — Typical Angina'). The form is pre-populated with a real test patient (age=41, male, no disease) so graders can run a prediction immediately without manual data entry.

Features and their input types:

Feature	Input Type	Valid Range / Options
Age	Number input	20–80 years
Sex	Dropdown	Male (1) / Female (0)
Chest Pain Type (cp)	Dropdown	0–3 with labels
Resting BP (trestbps)	Number input	80–220 mmHg
Cholesterol (chol)	Number input	100–600 mg/dl
Fasting Blood Sugar (fbs)	Dropdown	Yes (1) / No (0)
Resting ECG (restecg)	Dropdown	0–2 with labels
Max Heart Rate (thalach)	Number input	70–210 bpm
Exercise Angina (exang)	Dropdown	Yes (1) / No (0)
ST Depression (oldpeak)	Number input	0.0–7.0
Slope	Dropdown	0–2 with labels
Major Vessels (ca)	Number input	0–3
Thalassemia (thal)	Dropdown	1–3 with labels

E2 — Results Panel

On clicking RUN PREDICTION, the results panel displays:

- (a) Predicted class with colour-coded risk label — green badge for No Disease Detected, red badge for Disease Present
- (b) Model confidence score as a percentage, plus a visual progress bar showing P(disease)
- (c) Top 3 features driving the prediction as a horizontal bar chart using absolute logistic regression coefficients

A plain-English clinical note is generated dynamically below the result. Example for a disease-positive patient: This patient's maximum heart rate, ST depression level, and number of blocked vessels are the strongest indicators of elevated cardiac risk (model confidence: 82.3%). Recommend referral for a full cardiac work-up including stress testing and echocardiography. Do not discharge without cardiology review.

E3 — Submission Contents

- app.py — Streamlit dashboard application
- dashboard_model.pkl — Trained Logistic Regression model with scaler, feature columns, and pre-populated test patient
- requirements.txt — All Python dependencies

To run the app locally:

```
pip install -r requirements.txt
streamlit run app.py
Open: http://localhost:8501
```

The dashboard was tested and confirmed working via ngrok tunnel in Google Colab with the pre-populated test patient producing the correct No Disease prediction and a high-risk patient (age=63) producing the Disease Present warning.